



Relations : RACE-01

Single Correct Question

1. Let $X = \{1, 2, 3, 4, 5\}$ and $Y = \{1, 3, 5, 7, 9\}$. Which of the following is not a relation from X to Y
 (A) $R_1 = \{(x, y) | y = 2 + x, x \in X, y \in Y\}$
 (B) $R_2 = \{(1, 1), (2, 1), (3, 3), (4, 3), (5, 5)\}$
 (C) $R_3 = \{(1, 1), (1, 3), (3, 5), (3, 7), (5, 7)\}$
 (D) $R_4 = \{(1, 3), (2, 5), (2, 4), (7, 9)\}$
MRL108
2. If $A = \{2, 4, 5\}$, $B = \{7, 8, 9\}$, then $n(A \times B)$ is equal to
 (A) 6 (B) 9 (C) 3 (D) 0
MRL109
3. If $A = \{1, 2, 3\}$, $B = \{1, 4, 6, 9\}$ and R is a relation from A to B defined by ' x is greater than y ' ($x \in A, y \in B$). Then range of R is-
 (A) $\{1, 4, 6, 9\}$ (B) $\{4, 6, 9\}$ (C) $\{1\}$ (D) none of these
MRL110
4. The relation R is defined on the set of natural numbers as $\{(a, b) : a = 2b\}$. Then R^{-1} is given by
 (A) $\{(2, 1), (4, 2), (6, 3), \dots\}$ (B) $\{(1, 2), (2, 4), (3, 6), \dots\}$
 (C) R^{-1} is not defined (D) None of these
MRL111
5. The relation $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3)\}$ on set $A = \{1, 2, 3\}$ is
 (A) Reflexive but not symmetric (B) Reflexive but not transitive
 (C) Symmetric and Transitive (D) Neither symmetric nor transitive
MRL112
6. Let $P = \{(x, y) | x^2 + y^2 = 1, x, y \in R\}$. Then P is
 (A) Reflexive (B) Symmetric (C) Transitive (D) Anti-symmetric
MRL113
7. Let L be the set of all straight lines in the Euclidean plane. Two lines ℓ_1 and ℓ_2 are said to be related by the relation R if ℓ_1 is parallel to ℓ_2 . Then the relation R is-
 (A) Only Reflexive (B) Only Symmetric (C) Only Transitive (D) Equivalence
MRL114
8. Let X be a family of sets and R be a relation on X defined by ' A is disjoint from B '. Then R is-
 (A) reflexive (B) symmetric (C) anti-symmetric (D) transitive
MRL115

Relations

9. Let N denote the set of all natural numbers and R be the relation on $N \times N$ defined by $(a, b) R (c, d)$ if $ad(b + c) = bc(a + d)$, then R is-
- (A) Symmetric only (B) Reflexive only
(C) Transitive only (D) An equivalence relation

MRL116

10. **Statement-1** : Let $a, b \in I$

$R_1 \rightarrow aR_1b \Leftrightarrow a + b$ is an even integer then R_1 is an equivalence relation.

$R_2 \rightarrow aR_2b \Leftrightarrow a - b$ is an even integer, then R_2 is an equivalence relation.

Statement-2 : A relation which is reflexive, symmetric and transitive is called an equivalence relation.

- (A) Statement-1 is true, Statement-2 is true; Statement-2 is not the correct explanation of Statement-1
(B) Statement-1 is false, Statement-2 is true
(C) Statement-1 is true, Statement-2 is false
(D) Statement-1 is true, Statement-2 is true; Statement-2 is the correct explanation of Statement-1.

MRL117

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	B	C	B	A	B	D	B	D	D